MTH636 – Game Theory

Lecture 18: Winning Chomp & Chance Moves

1 Winning Strategies in Chomp

Recall Zermelo's Theorem: In any finite, perfect-information, two-player game with no chance moves, if there are no draws, either Player 1 or Player 2 must have a winning strategy. We apply this to David Gale's game of **Chomp**. The game cannot end in a draw, as someone must eventually eat the poisoned square $(1, 1)$. Therefore, a winning strategy exists. But who has it?

1.1 Square Boards ($n \times n$)

Theorem: Chomp on a Square Board

For an $n \times n$ board ($n > 1$), Player 1 has a winning strategy.

Proof. Player 1's winning strategy is explicit and constructive:

1. **Opening move:** Player 1 chooses the square $(2, 2)$. This removes a large block of squares and leaves a perfectly symmetrical board consisting of one row and one column of identical length meeting at the poisoned square $(1, 1)$.
2. **Subsequent moves:** Player 1 simply mirrors Player 2's moves. If Player 2 chooses a cell (i, j), Player 1 immediately chooses the symmetric cell (j, i).

Because Player 1 always restores symmetry, Player 2 is always faced with a symmetric board and is eventually forced to break the symmetry by eating the only square on the diagonal: $(1, 1)$. Thus, Player 1 wins. □

1.2 Rectangular Boards ($n \times m$)

Theorem: Chomp on a Rectangular Board

For any $n \times m$ board (where $n \neq m$), Player 1 *always* has a winning strategy.

Proof. We prove this by contradiction using a strategy-stealing argument. Because the game is finite and has no draws, Zermelo's theorem dictates that either Player 1 or Player 2 must have a winning strategy.

Suppose for contradiction that Player 2 has a winning strategy $\bar{s}_2$. Consider an experimental opening move by Player 1: Player 1 simply eats the single top-right corner square (n, m). Because Player 2 possesses a winning strategy $\bar{s}_2$, there must exist some specific response move (i, j) that Player 2 can make following Player 1's opening move, which guarantees a win for Player 2 from that point forward.

However, notice the geometry of Chomp: eating any square (i, j) automatically eats all squares above and to its right, including the top-right square (n, m). This means Player 1 could have simply skipped the "experimental" move and chosen (i, j) on their very first turn! The resulting board state would be exactly identical to the state created after Player 2's "winning" response.

If Player 1 does this, Player 1 effectively "steals" Player 2's winning strategy. Player 1 can now use $\bar{s}_2$ from that point forward to guarantee a win.

Thus, if Player 2 has a winning strategy, Player 1 can steal it to win, which is a logical contradiction (both players cannot have a winning strategy). Therefore, Player 2 cannot have a winning strategy. By Zermelo's theorem, Player 1 **must** have a winning strategy. □

Note: This is a classic non-constructive proof. We know Player 1 wins, but the proof does not tell us exactly what Player 1's opening move should be (unless the board is square).

2 Extensive Form Games with Chance

We now expand our model to include probabilistic elements (like rolling dice or drawing cards). We represent this by introducing a dummy player called **Nature** (Player 0), who moves randomly according to predetermined probabilities.

Definition: Extensive Form Game with Chance Moves

A game in extensive form with perfect information and chance moves is a tuple:

$$\Gamma = \langle N, V, E, n_0, (V_i)_{i \in N \cup \{0\}}, (p_n)_{n \in V_0}, (u_i)_{i \in N} \rangle$$

where:

- N is the set of strategic players.
- (V, E, n_0) is the directed rooted tree.
- $(V_i)_{i \in N \cup \{0\}}$ is a partition of the non-leaf nodes among the agents *and* Nature (P_0).
- For every node $n \in V_0$ (a node where Nature plays), p_n is a probability distribution over the outgoing edges $A(n)$ from n.
- $(u_i)_{i \in N}$ are the utility functions mapping leaf nodes to payoffs.

When the game reaches a node $n \in V_0$, no strategic choice is made. Instead, nature randomly selects a branch according to the explicit probabilities p_n.